

TECHNICAL & COMMERCIAL QUOTATION

OPTION 1 – DEDICATED DEHUMIDIFIER UNIT FOR EACH COLD ROOM

Quotation Ref.	MES/QTN/HVAC/ESM/2026-199	Date	27 August 2026
Client	M/s EBN SINA MEDICAL	Attention	Mr. Yasser Al Attar
Project	Desiccant Dehumidification of Three (3) Cold Rooms	Validity	30 days from date of offer

Subject: Supply & Installation of Desiccant Dehumidifiers for Three (3) Cold Rooms – Option 1: Individual Unit per Cold Room

Dear Mr. Yasser,

With reference to your kind enquiry and our subsequent discussion regarding humidity control inside your cold storage rooms, we are pleased to submit herewith our technical and commercial proposal for the supply and installation of desiccant (adsorption) dehumidifiers serving the three (3) cold rooms.

This offer covers Option 1, based on one dedicated dehumidifier unit installed for each cold room. A separate offer (Ref. MES/QTN/DEH/ESM/2026-206A) is submitted in parallel covering Option 2, based on a single centralised dehumidifier serving all three rooms.

1. PROPOSED SOLUTION

We propose to supply and install three (3) nos. AIRDRY model ADS 450 adsorption dehumidifiers, manufactured by Tecnofrigo Tuscany Srl (TFT), Italy — one dedicated unit for each cold room.

- Each unit is installed outside or inside its respective cold room, in the adjacent ambient area, mounted on a wall or ceiling suspended.
- Process air is drawn from the cold room, dried across the silica-gel rotor, and the dry air is ducted back into the room through an insulated duct connection.
- The regeneration (wet) air stream is ducted and discharged to atmosphere, outside the treated space.
- Each room is provided with its own wall-mounted mechanical humidistat, so that the relative humidity of each cold room can be set and controlled independently.

Advantages of this option: full independence between rooms, individual RH set-points, no cross-connection of air between cold rooms, single-phase power supply only, minimal ductwork, and continuity of operation in the other two rooms should one unit be taken out of service for maintenance.

2. SCOPE OF WORK

No.	Description
1	Supply & installation of 3 nos. AIRDRY ADS 450 adsorption dehumidifiers (TFT – Italy), each complete with one ADKMH2 mechanical wall-mounted humidistat.
2	Testing, commissioning, setting of the required RH set-point and verification of performance in each room.

3. TECHNICAL PARTICULARS – AIRDRY ADS 450

EQUIPMENT IDENTIFICATION	
Equipment	Adsorption (desiccant) dehumidifier, silica-gel rotor type
Model	AIRDRY ADS 450
Brand / Manufacturer	TFT – Tecnofrigo Tuscany Srl, Capanne (Pisa), Italy
Country of Origin	Italy
Quantity	3 Nos. (one per cold room)
PERFORMANCE	
Dehumidification Capacity	1.62 kg/h per unit at 20 °C / 60% RH (total 4.86 kg/h for 3 units)
Process Air Flow	450 m ³ /h per unit
Process Operating Temperature Range	–30 °C to +40 °C
ELECTRICAL	
Power Supply	230 V / 1 Phase + N / 50 Hz ±5%
Installed Regeneration Heater Power	1.40 kW per unit
Process Fan Nominal Power	157 W per unit
Rotor Drive Motor	3.7 VA
Maximum Power Absorbed	1.56 kW per unit
Maximum Current Absorbed	5.4 A per unit
DIMENSIONS, WEIGHT & CONNECTIONS	
Dimensions (A × B × C)	424 × 455 × 432 mm
Empty Weight	21.6 kg per unit (total 64.8 kg)
Process / Regeneration Air Inlet	397 × 233 mm
Dry Air Outlet	Ø 160 mm
Wet (Regeneration) Air Outlet	Ø 63 mm
CONTROL ACCESSORY	
Humidistat	ADKMH2 – mechanical wall-mounted humidistat, control range 30 – 90% RH, 1 step, IP54 (3 Nos. included, one per room)

4. COMMERCIAL OFFER

No.	Description	Qty	Unit	Rate (QAR)	Amount (QAR)
1	Supply, delivery and installation of AIRDRY ADS 450 adsorption dehumidifier (TFT – Italy), complete with ADKMH2 mechanical humidistat, mounting bracket, insulated dry air and process air duct connections, wet air discharge duct with weather louvre, control wiring, final connection, testing, commissioning and handover — all as per the Scope of Work and Technical Particulars above.	3	Nos.	33,000.00	99,000.00
TOTAL LUMP SUM PRICE – OPTION 1					99,000.00

5. COMMERCIAL TERMS & CONDITIONS

Payment Terms	70% advance payment against Purchase Order 30% against completion of installation and commissioning
Delivery Period	3 – 4 weeks from date of receipt of the advance payment
Installation Period	Within 5 – 7 working days from date of delivery to site, subject to site readiness
Warranty	12 months manufacturer's warranty on the equipment against manufacturing defects from the date of delivery, plus 12 months warranty on our installation workmanship.
Validity of Offer	30 (thirty) days from the date of this quotation
Currency	Qatari Riyals (QAR). Prices are for supply and installation at site, Doha – State of Qatar

We trust the above meets with your kind approval and remain at your disposal for any technical clarification or for a joint site visit to finalise the installation arrangement.

We thank you for the opportunity to quote and look forward to your valued order.

Yours faithfully,

Metri Engineering Services W.L.L.



AIRDRY

Adsorption Dehumidifiers

ADS 150÷450



TECHNICAL DATA

MODEL	ADS	150	250	450
Performances				
Dehumidification Capacity *	Kg/h	0,47	0,78	1,62
Fans				
Process air flow	m ³ /h	150	250	450
Static pressure	Pa	50	35	110
Fan nominal power	W	85	71	157
Regeneration air flow	m ³ /h	50	60	120
Static pressure	Pa	40	30	90
Fan nominal power	W	-	-	-
Drive Motor				
Nominal power	VA	3,7	3,7	3,7
Regeneration				
Regeneration type		Electric	Electric	Electric
Installed power	KW	0,7	1,4	1,4
Temperature rise in the heating coil	°C	70	90	90
Electrical characteristics				
Power supply	Volt/Ph/Hz	230/1+N/50 ±5%	230/1+N/50 ±5%	230/1+N/50 ±5%
Maximum power absorbed standard units	KW	0,785	1,47	1,56
Maximum absorbed standard units	A	2,8	4,1	5,4
Noise level				
Sound pressure **	dB (A)	42	48	55
Sound power **	dB (A)	70	76	83

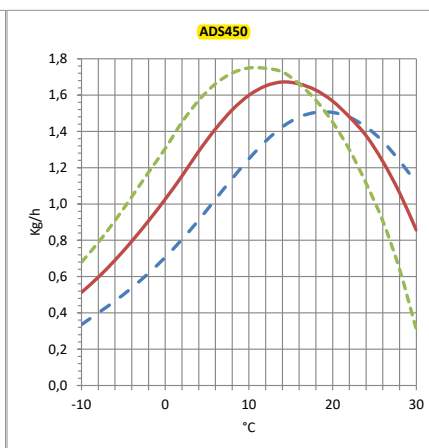
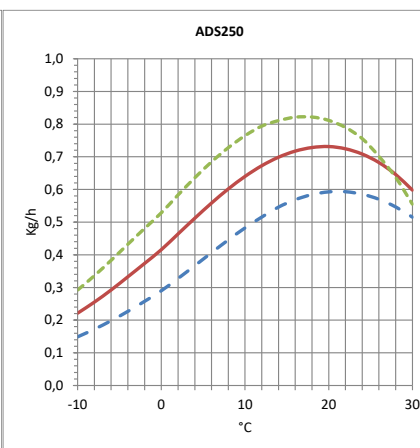
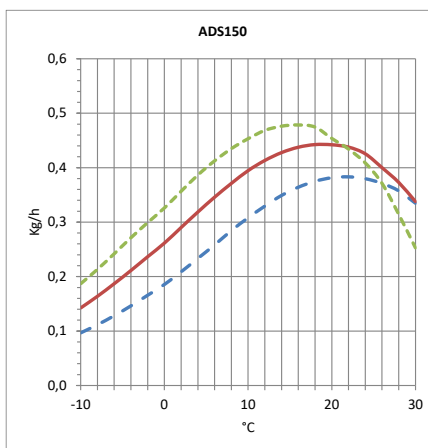
* At the conditions of 20°C 60% RH

** Sound pressure level calculated in free field, 1 meter from the unit, directionality factor Q=2, according to ISO 9614

DEHUMIDIFICATION CAPACITY

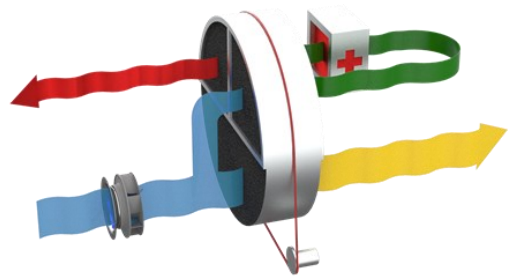
Approximate capacity in kg/h with different values of relative humidity of inlet process air (%RH).

— 40% RH — 60% RH — 80% RH



PRINCIPLE OF OPERATION

The dehumidifier operates using two streams of air; the main one is the air to be dehumidified, while a second stream - of a smaller flow rate - is used to regenerate the dehumidification rotor. A fan inside the dehumidifier creates these two streams of air that pass through the rotor in opposite directions. The air to be dehumidified-also called "process air"-runs through the desiccant rotor impregnated with silica gel. Silica gel is a highly hygroscopic material that absorbs water vapor from the air. As it passes through the rotor, the air gives up its moisture content to the rotor. The dehumidified air is then sent into the production room or process to be dehumidified. The dehumidification process can take place between temperatures of -30°C and +40°C. During the process, the rotor turns very slowly and is equipped with a transmission system with a reducer motion and belt. The so-called "regeneration air," is used by the system to remove the absorbed moisture and carry it outside: it is heated by means of a battery inside the dehumidifier, up to about +100°C and passes through the rotor in the opposite direction from the process air and subjects it to a reverse process, whereby the rotor gives up its moisture content and its initial absorptive capacity is restored. The regeneration air is expelled warm and moist and must be sent outside the treated environment.



STRUCTURE

The structure of the dehumidifier is made of galvanized steel and steel AISI_304. The top panel can be removed for maintenance on electrical components and all other internal mechanical parts. The dehumidifier connections can be made with standard spiral ducts.

FANS

Fans are directly coupled to single-phase or three-phase motors rated at IP55, ISO F, class B. They are accessible for maintenance by removing the inspected top panel. The process and regenerator fans immediately start to dehumidify the system.

ROTOR

The dehumidifier has a rotor made of desiccant material. The rotor has a honeycomb structure made of corrugated, heat-resistant sheets containing the desiccant material silica gel, which creates a high number of axial fluid threads while creating a high absorption surface area in a small volume. The rotor is constructed to withstand saturated air without being damaged. In addition, the rotor is not damaged if the process or regeneration fan should stop due to failure during operation. The rotor is non-combustible and non-flammable.

TRANSMISSION SYSTEM

A belt drive system is used to turn the rotor. This movement is typically between 6 and 12 rph, and uses a powerful direct drive motor and reduction gearbox, operating on a belt with frictional contact with the outer rim of the rotor drum. A belt tensioning system is used to maintain correct belt tension and avoid slip. The rotation of the rotor is visible by removal of the front access panel so correct operation can be determined. The rotor is suspended on ball bearings around a central steel shaft.

REGENERATION AIR HEATING COIL

Electrical. The electric regeneration coil is of the self-regulating PTC type to keep the surface temperature constant.

FILTERS

The dehumidifier has two separate G2 filters: on the process air and regeneration air inlet.

ELECTRICAL PANEL

The electrical panel is made in adherence to European standards 73/23 and 89/336. The electrical panel can be accessed by removing the top panel of the unit. The following components are installed in all units as standard: main switch, ammeter, hour meter, connector for external humidistat connection. The panel is also equipped with a switch for manual or automatic control of dehumidification management.

INSTALLATION NOTES

The ADS series units do not have thermal insulation, so during selection and/or installation it is necessary to consider the various climatic conditions between process (indoor) air and outdoor air, to avoid condensation or ice formation on the panels, and possible damage to electrical components.

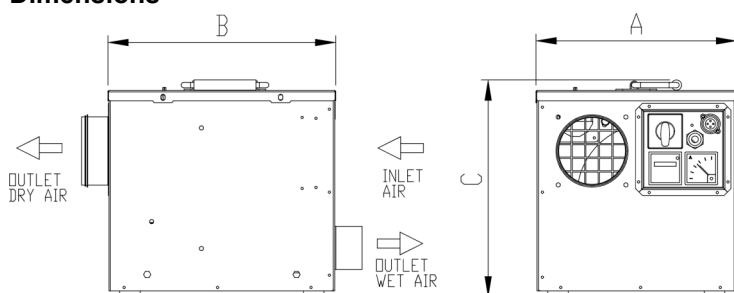
VERSION

ADS...	Standard
ADS.../TRLL	Version with pivoting wheels

Model ADS	Code	150	250	450
Painted steel frame		-	-	-
Satin stainless steel frame 304		●	●	●
Structure with pivoting wheels	TRLL	○	○	○
Main power switch		●	●	●
Operating hour meter		●	●	●
Ammeter		●	●	●
Frame Handed version	M	-	-	-
Cover for outdoor installation	ADKOPB	○	○	○
Filters G2 process and reactivation		●	●	●
Filters F5, F7, F9		-	-	-
Electronic PLC control and touch-screen terminal		-	-	-
Different power supply voltage		○	○	○
Process filter alarm	ALFP	-	-	-
Regeneration filter alarm	ALFR	-	-	-
Mechanical humidistat for duct or wall 10÷100% IP54 1 step	ADKMH1	○	○	○
Mechanical humidistat for wall 30÷90% IP54 1 step	ADKMH2	○	○	○
Electronic wall-mounted 1-step humidistat with channel (D) or wall (W) probe temperature/relative humidity range -30÷70°C / 0÷100%	ADKW230 + ADKH1D/W	○	○	○
Electronic wall-mounted 1-step humidistat with built-in temperature/relative humidity probe 10÷90%	ADKHTW1	○	○	○

● standard, ○ optional, – not available.

Dimensions



Model	ADS	150	250	450
A	mm	295	295	424
B	mm	335	352	455
C	mm	318	338	432
Empty weight	Kg	11,0	11,3	21,6
Connections				
Process and regeneration air inlet	mm	267 x 138	267 x 138	397 x 233
Dry air outlet	mm	Ø 100	Ø 100	Ø 160
Wet air outlet	mm	Ø 63	Ø 63	Ø 63